

An Analysis of Deep Q-network

Deep Q-network

Deep Q-network -- Part 1

Receptive field In neural networks, each neuron receives input from some number of locations in the previous layer. In a convolutional layer, each neuron receives input from only a restricted area of the previous layer called the neuron's receptive field. Typically the area is a square (e.g. 5 by 5 neurons). Whereas, in a fully connected layer, the receptive field is the entire previous layer. Thus, in each convolutional layer, each neuron takes input from a larger area in the input than previous layers. This is due to applying the convolution over and over, which takes the value of a pixel into account, as well as its surrounding pixels. When using dilated layers, the number of pixels in the receptive field remains constant, but the field is more sparsely populated as its dimensions grow when combining the effect of several layers. To manipulate the receptive field size as desired, there are some alternatives to the standard convolutional layer. For example, atrous or dilated convolution expands the receptive field size without increasing the number of parameters by interleaving visible and blind regions. Moreover, a single dilated convolutional layer can comprise filters with multiple dilation ratios, thus having a variable receptive field size.

Deep Q-network -- Part 2

Number of filters Since feature map size decreases with depth, layers near the input layer tend to have fewer filters while higher layers can have more. To equalize computation at each layer, the product of feature values v_a with pixel position is kept roughly constant across layers. Preserving more information about the input would require keeping the total number of activations (number of feature maps times number of pixel positions) non-decreasing from one layer to the next. The number of feature maps directly controls the capacity and depends on the number of available examples and task complexity.

Deep Q-network -- Part 3

Max norm constraints Another form of regularization is to enforce an absolute upper bound on the magnitude of the weight vector for every neuron and use projected gradient descent to enforce the constraint. In practice, this corresponds to performing the parameter update as normal, and then enforcing the constraint by clamping the weight vector $\mathbf{w} \rightarrow \{\vec{w}\}$ of every neuron to satisfy $\|\mathbf{w}\|_2 < c$. Typical values of c are order of 3–4. Some papers report improvements when using this form of regularization.

Deep Q-network -- Part 4

Image recognition with CNNs trained by gradient descent Denker et al. (1989) designed a 2-D CNN system to recognize hand-written ZIP Code numbers. However, the lack of an efficient training method to determine the kernel coefficients of the involved convolutions meant that all the coefficients had to be laboriously hand-designed. Following the advances in the training of 1-D CNNs by Waibel et al. (1987), Yann LeCun et al. (1989) used back-propagation to learn the convolution kernel coefficients directly from images of hand-written numbers. Learning was thus fully automatic, performed better than manual coefficient design, and was suited to a broader range of image recognition problems and image types. Wei Zhang et al. (1988) used back-propagation to train the convolution kernels of a CNN for alphabets recognition. The model was called shift-invariant pattern recognition neural network before the name CNN was coined later in the early 1990s. Wei Zhang et al. also applied the same CNN without the last fully connected layer for medical image object segmentation (1991) and breast cancer detection in mammograms (1994). This approach became a foundation of modern computer vision.

Deep Q-network -- Part 5

"S-layer": a shared-weights receptive-field layer, later known as a convolutional layer, which contains units whose receptive fields cover a patch of the previous layer. A shared-weights receptive-field group (a "plane" in neocognitron terminology) is often called a filter, and a layer typically has several such filters. **"C-layer"**: a downsampling layer that contain units whose receptive fields cover patches of previous convolutional layers. Such a unit typically computes a weighted average of the activations of the units in its patch, and applies inhibition (divisive normalization) pooled from a somewhat larger patch and across different filters in a layer, and applies a saturating activation function. The patch weights are nonnegative and are not trainable in the original neocognitron. The downsampling and competitive inhibition help to classify features and objects in visual scenes even when the objects are shifted. Several supervised and unsupervised learning algorithms have been proposed over the decades to train the weights of a neocognitron. Today, however, the CNN architecture is usually trained through backpropagation. Fukushima's ReLU activation function was not used in his neocognitron since all the weights were nonnegative; lateral inhibition was used instead. The rectifier has become a very popular activation function for CNNs and deep neural networks in general.

Deep Q-network -- Part 6

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Time delay neural networks The time delay neural network (TDNN) was introduced in 1987 by Alex Waibel et al. for phoneme recognition and was an early convolutional network exhibiting shift-invariance. A TDNN is a 1-D convolutional neural net where the convolution is performed along the time axis of the data. It is the first CNN utilizing weight sharing in combination with a training by gradient descent, using backpropagation. Thus, while also using a pyramidal structure as in the neocognitron, it performed a global optimization of the weights instead of a local one. TDNNs are convolutional networks that share weights along the temporal dimension. They allow speech signals to be processed time-invariantly. In 1990 Hampshire and Waibel introduced a variant that performs a two-dimensional convolution. Since these TDNNs operated on spectrograms, the resulting phoneme recognition system was invariant to both time and frequency shifts, as with images processed by a neocognitron. TDNNs improved the performance of far-distance speech recognition.

Deep Q-network -- Part 7

Distinguishing features In the past, traditional multilayer perceptron (MLP) models were used for image recognition. However, the full connectivity between nodes caused the curse of dimensionality, and was computationally intractable with higher-resolution images. A 1000×1000-pixel image with RGB color channels has 3 million weights per fully-connected neuron, which is too high to feasibly process efficiently at scale.

Deep Q-network -- Part 8

The "loss layer", or "loss function", exemplifies how training penalizes the deviation between the predicted output of the network, and the true data labels (during supervised learning). Various loss functions can be used, depending on the specific task. The Softmax loss function is used for predicting a single class of K mutually exclusive classes. Sigmoid cross-entropy loss is used for predicting K independent probability values in $[0, 1]$ $\{ \displaystyle [0,1] \}$. Euclidean loss is used for regressing to real-valued labels $(-\infty, \infty)$ $\{ \displaystyle (-\infty, \infty) \}$.